

Bandwidth controls optimization access to high precision

Section 3.4 working note

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Proposed subsection, followed by its full spectral argument and evaluation details. References to Sections 3.1–3.3 and Figure 2b refer to submission draft (19). The final paragraph leads into the forthcoming joint-feature analysis.

3.4 BANDWIDTH CONTROLS OPTIMIZATION ACCESS TO HIGH PRECISION

The useful relative bandwidth $\lambda = \gamma h$ varies slowly with width once the target is resolved (Figure 2b). Maintaining this bandwidth as $h = \Theta(W^{-1})$ therefore requires $\gamma = \Theta(W)$. Even though larger widths admit smaller λ at comparable construction error, small slopes can still delay readout learning by weakening target-relevant kernel directions. We quantify this effect first with fixed features, then examine whether joint training acquires useful feature geometry.

Consider the construction of Section 3.1, but solve its normalized readout least-squares problem by gradient descent. The sampled feature matrix Φ_γ includes the output bias; its kernel is $K_\gamma = \Phi_\gamma \Phi_\gamma^\top / M$, where M is the number of training points. If its eigenvalues are $\mu_1 \geq \dots \geq \mu_M \geq 0$, define

$$\rho_i(\gamma) = \frac{\mu_i(K_\gamma)}{\mu_1(K_\gamma)}. \quad (1)$$

The largest eigenvalue limits a stable shared step size; a small ρ_i then makes the corresponding output component slow to learn.

How slope reaches these rates. A tanh feature changes over distance $1/\gamma$. At small slope, broad, overlapping transitions respond weakly to rapidly alternating errors. After removing the common constant component, the center-integrated kernel is determined by the explicit pairwise profile

$$-d \coth(\gamma d) = -|d| - \frac{2|d|}{e^{2\gamma|d|} - 1}, \quad d = z_a - z_b, \quad (2)$$

with value $-1/\gamma$ at $d = 0$. Here z_a, z_b are any two training points. The second term is the slope-dependent smoothing correction to the step-feature profile $-|d|$. Bounding the discrepancy from the actual finite center lattice transfers this explicit gamma dependence to the kernel spectrum.

Theorem 3.2 (Slope-dependent relative kernel spectrum). *For the uniformly spaced centers and training points of Section 3.1 and a common slope $\gamma > 0$, the correlation profile (2) yields computable bounds*

$$\boxed{\underline{\rho}_i(\gamma) \leq \frac{\mu_i(K_\gamma)}{\mu_1(K_\gamma)} \leq \bar{\rho}_i(\gamma), \quad i = 1, \dots, M.} \quad (3)$$

The endpoints depend only on the slope and fixed feature geometry, and lie in $[0, 1]$. Their explicit construction, finite-lattice and halo allowances, and proof are given in Appendix A.

Figure 3 makes this dependence quantitative: raising λ from $1/32$ to $1/4$ changes the fourth relative eigenvalue by only a factor of 1.30, but the thirtieth by 1.4×10^5 . The bounds follow this change by computing the spectrum of the explicit, corrected correlation model before optimization. Whether these slow directions matter for training depends on the target’s projections.

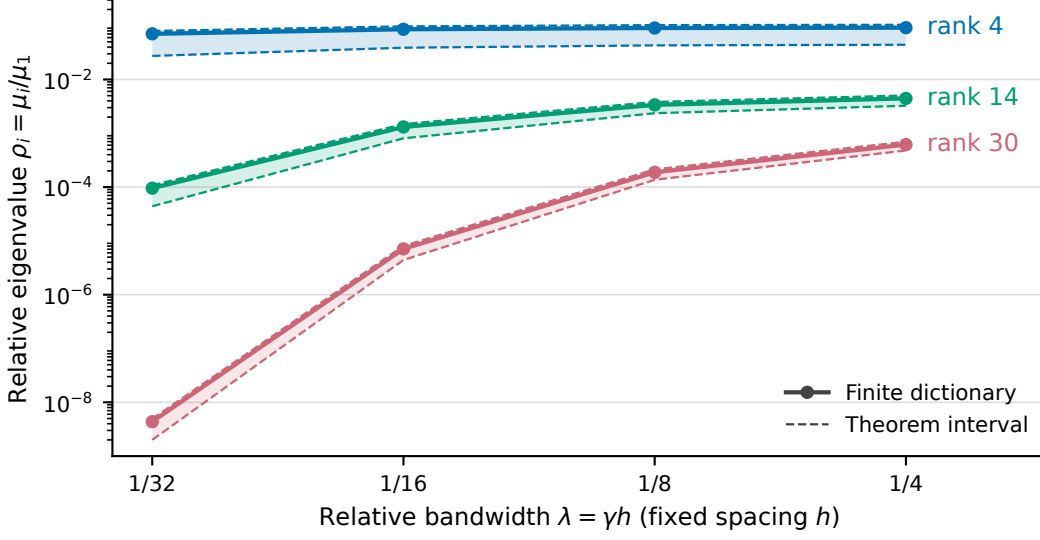


Figure 3: **Slope changes relative rates by very different amounts.** Finite-kernel ratios (solid, dots) and theorem intervals (dashed boundaries, shading) for three illustrative ordered ranks, at fixed width 512. Four bandwidths are evaluated; connecting lines are guides. Rank 30 carries 2.6% and 2.5% of target energy at the two endpoints, while its relative rate increases by 1.4×10^5 . Ordered rank does not identify the same eigenvector across slopes. All figures use the target and geometry in Appendix B.

Corollary 3.3 (Output error from slow target components). *For sampled target $y \neq 0$, let $p_i(\gamma) = |u_i^\top y|^2 / \|y\|^2$ be its energy fraction in kernel eigenvector u_i . For zero-initialized readout GD with $\eta = 1/(2\mu_1)$, the relative output error satisfies*

$$e_n^2 := \frac{\|\Phi_\gamma w_n - y\|^2}{\|y\|^2} \geq \sum_i p_i(\gamma) \left(1 - \frac{\bar{p}_i(\gamma)}{2}\right)^{2n}. \quad (4)$$

Indeed, the exact GD error is the same sum with ρ_i in place of $\bar{p}_i$. If directions carrying target energy P have $\bar{p}_i \leq r$, they force $e_n \geq \sqrt{P}(1 - r/2)^n$. Small positive eigenvalues therefore delay components already representable by the dictionary; they need not create a nonzero approximation floor.

Accurate readouts can remain slow to acquire. At $\lambda = 1/32$, a direct readout fit achieves 6.2×10^{-10} relative error on a dense evaluation grid. Yet GD retains training error 0.210 after five million updates; the corollary predicts at least 0.2098 (Figure 4A). At $\lambda = 1/4$, GD reaches 3.88×10^{-4} over the same budget. Across the four bandwidths, the lower bound is at least 75% of executed error throughout training and 91–100% at the endpoint. Thus the gamma-dependent spectral bounds quantify an optimization obstruction beyond attainable approximation error.

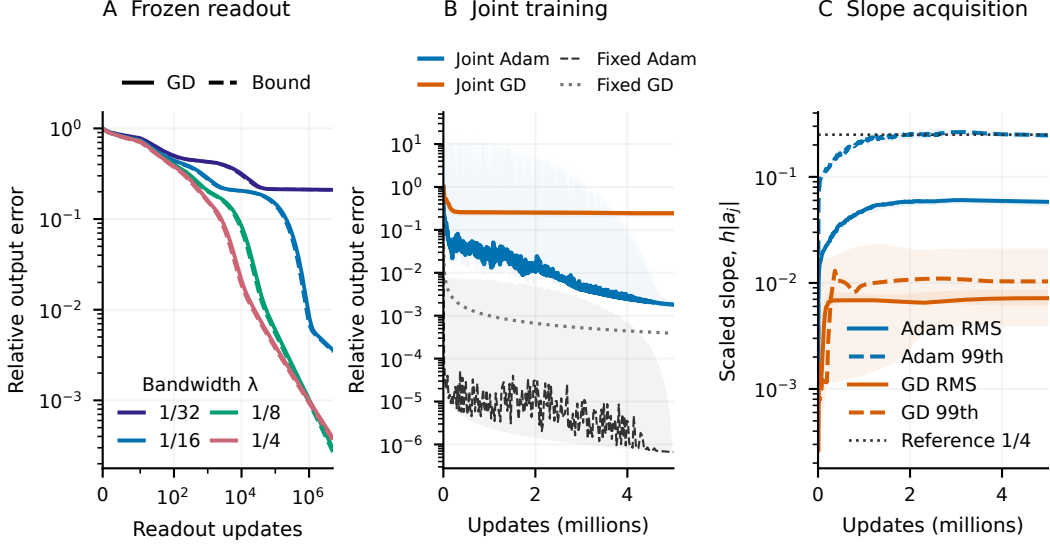


Figure 4: **From readout delay to feature acquisition.** **A:** executed frozen-readout GD errors (solid) and the square roots of the corollary’s lower bounds (dashed). **B:** joint Adam/GD errors, with frozen uniform $\lambda = 1/4$ references over the same five-million-update budget. **C:** RMS and 99th-percentile scaled slopes $h|a_j|$ from B’s joint runs. The reference bandwidth is a supplied geometry, not a necessary threshold for heterogeneous features. Joint curves summarize five paired seeds; shading retains seed variation and recorded within-bin extrema. The full horizon is shown and cosine schedules span that horizon. Appendix B specifies the error measure and selection protocol. The theorem applies to A; B–C are empirical.

Joint training must acquire useful slopes and centers together. Joint Adam reaches median relative error 1.81×10^{-3} , whereas Adam on the supplied $\lambda = 1/4$ features reaches 6.62×10^{-7} over the same budget; joint GD remains at 0.244 (Figure 4B). Adam’s upper slope percentile approaches the reference scale, but only 4–7 of 512 features reach it (Figure 4C). A few steep features do not establish that the useful geometry has been acquired. Moreover, refitting the learned readout barely improves its error, while steepening the learned features alone can worsen it. These observations motivate the next question: what training signal moves slopes and centers into a jointly useful configuration? The next analysis addresses this feature-acquisition step; the frozen theorem isolates the readout obstruction once geometry is fixed.

The construction supplies both feature geometry and accurate readout weights, avoiding these two optimization tasks. This motivates embedding the resulting arithmetic primitives in larger networks. The underlying scale-dependent spectral bias is consistent with prior analysis (Xu et al., 2020; Tancik et al., 2020); here its consequences are quantified for the finite quasi-interpolant dictionary.

A Explicit spectral bounds and proof

The main theorem concerns the actual finite tanh dictionary. An auxiliary center integral exposes its gamma dependence; lattice and exterior-center allowances transfer its spectrum to that dictionary. This appendix specifies each approximation and the resulting endpoints.

Finite setup. Let $z_1, \dots, z_M$ be distinct training points with $M \geq 2$, W consecutive centers on $c_0 + h\mathbb{Z}$ cover their interval, and $\gamma > 0$. Define

$$\Phi_\gamma = [\mathbf{1}, (\tanh(\gamma(z_a - x_j)))_{a,j}], \quad K_\gamma = \frac{1}{M} \Phi_\gamma \Phi_\gamma^\top, \quad y = (f(z_a))_{a=1}^M \neq 0. \quad (5)$$

Readout GD minimizes $\|\Phi_\gamma w - y\|^2/(2M)$. The theorem permits arbitrary distinct samples; the reported experiments use uniform samples. Write $v_0 = \mathbf{1}/\sqrt{M}$ and let Q have orthonormal columns spanning $v_0^\perp$. This projection removes the common constant sample pattern, not any target assumption. Put $t_\gamma(c) = (\tanh(\gamma(z_a - c)))_a$.

The gamma-dependent correlation model. Its center-integrated, mean-zero kernel is

$$H_\gamma = \frac{1}{hM} \int_{\mathbb{R}} Q^\top t_\gamma(c) t_\gamma(c)^\top Q dc = -\frac{2}{hM} Q^\top [d_{ab} \coth(\gamma d_{ab})]_{a,b} Q, \quad d_{ab} = z_a - z_b. \quad (6)$$

The diagonal of the bracketed matrix is $1/\gamma$. Choose a finite set $\mathcal{F}$ of exterior lattice centers absent from the actual dictionary, leaving two unretained tails starting at $c_L < \min z_a$ and $c_R > \max z_a$. Subtract their retained contribution:

$$T_\gamma = \frac{1}{M} \sum_{c \in \mathcal{F}} Q^\top t_\gamma(c) t_\gamma(c)^\top Q, \quad S_\gamma = H_\gamma - T_\gamma. \quad (7)$$

Let $\beta_1 \geq \dots \geq \beta_{M-1}$ be the eigenvalues of S_γ . For any $0 < \theta < \pi/2$, the lattice and remaining-tail allowances are

$$\delta_\gamma = \frac{8\gamma \|z - \bar{z}\mathbf{1}\|^2 \sec^4 \theta}{3hM [\exp(2\pi\theta/(\gamma h)) - 1]}, \quad \bar{z} = M^{-1} \sum_a z_a, \quad (8)$$

$$\tau_\gamma = \frac{4[e^{-4\gamma(c_R - z_{\max})} + e^{-4\gamma(z_{\min} - c_L)}]}{1 - e^{-4\gamma h}}. \quad (9)$$

They give the matrix enclosure

$$S_\gamma - (\delta_\gamma + \tau_\gamma)I \preceq Q^\top K_\gamma Q \preceq S_\gamma + \delta_\gamma I. \quad (10)$$

Relative-eigenvalue endpoints. Normalize using scalar bounds $\ell_\gamma \leq \mu_1 \leq L_\gamma$. The following require only matrix-vector products with the finite dictionary:

$$\ell_\gamma = v_0^\top K_\gamma v_0 \geq 1, \quad b_\gamma = \|Q^\top K_\gamma v_0\|, \quad c_\gamma = \beta_1 + \delta_\gamma, \quad (11)$$

$$L_\gamma = \frac{\ell_\gamma + c_\gamma + \sqrt{(\ell_\gamma - c_\gamma)^2 + 4b_\gamma^2}}{2}. \quad (12)$$

Set $\rho_1 = \bar{\rho}_1 = 1$ and $\beta_M = 0$. The endpoints in Theorem 3.2 are

$$\boxed{\rho_i = \frac{[\beta_i - \delta_\gamma - \tau_\gamma]_+}{L_\gamma}, \quad \bar{\rho}_i = \min \left\{ 1, \frac{\beta_{i-1} + \delta_\gamma}{\ell_\gamma} \right\}, \quad 2 \leq i \leq M.} \quad (13)$$

Exact zeros known from $\text{rank } K_\gamma \leq W + 1$ may have both endpoints set to zero. No target or optimization trajectory enters these spectral bounds.

A.1 DERIVATION OF THE EXPLICIT GAMMA DEPENDENCE

The identity $1 - \tanh u \tanh v = \coth(u - v)(\tanh u - \tanh v)$ gives

$$\int_{\mathbb{R}} [1 - \tanh(\gamma(z_a - c)) \tanh(\gamma(z_b - c))] dc = 2d_{ab} \coth(\gamma d_{ab}). \quad (14)$$

Multiplication by $Q^\top$ and Q removes the constant term and proves (6). The integral is finite after projection because $Q^\top t_\gamma(c)$ decays at both ends.

The smoothing interpretation can also be made exact. With $s_\gamma(x) = \frac{\gamma}{2} \operatorname{sech}^2(\gamma x)$, $\tanh(\gamma \cdot) = s_\gamma * \operatorname{sign}$. Its Fourier multiplier is

$$A_\gamma(\omega) = \widehat{s}_\gamma(\omega) = \frac{\pi|\omega|/(2\gamma)}{\sinh(\pi|\omega|/(2\gamma))}, \quad A_\gamma(0) = 1. \quad (15)$$

For completeness, substitution $r = e^{2\gamma x}$ in the Fourier transform gives $\int_0^\infty r^{-i\omega/(2\gamma)}(1+r)^{-2} dr$. The beta integral evaluates this as $\Gamma(1 - i\omega/(2\gamma))\Gamma(1 + i\omega/(2\gamma))$, which equals (15) by the reflection formula. For $u = Qv$, the step combination $\sum_a u_a \operatorname{sign}(c - z_a)$ has compact support, and its derivative is $2 \sum_a u_a \delta_{z_a}$. Parseval gives

$$v^\top H_\gamma v = \frac{2}{\pi h M} \int_{\mathbb{R}} \frac{A_\gamma(\omega)^2}{\omega^2} \left| \sum_a (Qv)_a e^{-i\omega z_a} \right|^2 d\omega. \quad (16)$$

The apparent zero-frequency singularity is removable since $\sum_a (Qv)_a = 0$. Thus gamma acts on a direction's frequency content through the explicit weight A_γ^2 . Since $t/\sinh t$ decreases for $t > 0$, increasing gamma increases every directional response of H_γ and hence its ordered eigenvalues. This order concerns the integral kernel; exterior subtraction and normalization need not preserve monotonicity of every finite ratio.

A.2 TRANSFER TO THE FINITE DICTIONARY

For $u \perp \mathbf{1}$, the function $g(\zeta) = \sum_a u_a \tanh(\gamma(\zeta - z_a))$ is analytic for $|\operatorname{Im} \zeta| < \pi/(2\gamma)$. Subtracting the common translate at $\bar{z}$ and integrating derivatives, Minkowski and Cauchy-Schwarz yield

$$\int_{\mathbb{R}} |g(t \pm i\theta/\gamma)|^2 dt \leq \frac{4\gamma}{3} \sec^4 \theta \|z - \bar{z}\mathbf{1}\|^2 \|u\|^2. \quad (17)$$

Here $|\operatorname{sech}^2(t + i\theta)| \leq \sec^2 \theta \operatorname{sech}^2 t$ and $\int \operatorname{sech}^4 t dt = 4/3$. Shift the contour for the analytic function $g(\zeta)^2$, not $|g(\zeta)|^2$: its transform is bounded by the right side of (17) times $e^{-\theta|\omega|/\gamma}$. Poisson summation at frequencies $2\pi k/h$ then gives

$$\left\| \frac{1}{M} \sum_{c \in c_0 + h\mathbb{Z}} Q^\top t_\gamma(c) t_\gamma(c)^\top Q - H_\gamma \right\| \leq \delta_\gamma. \quad (18)$$

The geometric sum over $k \neq 0$ produces the denominator in (8).

Deleting exterior centers converts the infinite lattice to the actual dictionary. The retained deletions give T_γ . For a remaining center $c \geq c_R$, projection removes the constant limit and $1 - \tanh t \leq 2e^{-2t}$ gives an outer-product norm, divided by M , of at most $4e^{-4\gamma(c - z_{\max})}$. Summing this geometric series and its left-tail counterpart bounds the remaining positive deletion by $\tau_\gamma I$. This proves (10).

If κ_i are the eigenvalues of $Q^\top K_\gamma Q$, min-max and codimension-one interlacing give

$$\beta_i - \delta_\gamma - \tau_\gamma \leq \kappa_i \leq \beta_i + \delta_\gamma, \quad \mu_i \geq \kappa_i \geq \mu_{i+1}. \quad (19)$$

The Rayleigh quotient gives $\mu_1 \geq \ell_\gamma$. In the decomposition $\operatorname{span}(v_0) \oplus v_0^\perp$, the diagonal blocks of K_γ are bounded above by ℓ_γ and $c_\gamma I$, and the off-diagonal block has norm b_γ . Its quadratic form is therefore bounded by the largest eigenvalue of $\begin{bmatrix} \ell_\gamma & b_\gamma \\ b_\gamma & c_\gamma \end{bmatrix}$, namely L_γ . Dividing the interlacing bounds by these normalization endpoints proves (13) and the main theorem.

A.3 CLASSICAL OUTPUT-ERROR CONSEQUENCE

For $r_n = \Phi_\gamma w_n - y$, normalized least-squares GD gives $r_{n+1} = (I - \eta K_\gamma) r_n$ and $r_0 = -y$. Expansion in the actual finite-kernel eigenvectors gives

$$e_n^2 = \sum_i p_i (1 - \eta \mu_i)^{2n} = \sum_i p_i (1 - \rho_i/2)^{2n} \geq \sum_i p_i (1 - \bar{\rho}_i/2)^{2n}. \quad (20)$$

The last inequality follows because the factor decreases with $\rho_i \in [0, 1]$. This proves Corollary 3.3. Our step is a conservative fraction of the stability limit $2/\mu_1$; any nonoscillatory step $0 < \eta \mu_1 \leq 1$ instead uses $\eta \mu_1 \rho_i$. Only exact nullspace energy gives a permanent floor. For $\mu_i > 0$, $u_i = \Phi_\gamma(\Phi_\gamma^\top u_i / (M \mu_i))$ is representable, however small its rate. The target weights require the actual eigenvectors; the theorem bounds eigenvalues, not their target alignment.

B Evaluation, accuracy, and empirical scope

Which quantities are computed. The theorem endpoints use the explicit center integral, exterior-center subtraction, and scalar normalization bounds. A separate SVD of the actual finite feature matrix supplies the validation eigenvalues and target weights p_i . The plotted lower error uses the theorem’s upper rates, not actual rates. The spectral figure selects three ranks for readability; the error bound includes every numerically resolved target projection.

The integral is evaluated through a positive quadrature factor rather than by subtracting nearly equal entries of (6). There is no Fourier truncation. Truncating the center integral at $\pm(\max_a |z_a| + P/\gamma)$ omits a positive matrix of norm at most $2e^{-4P}/(h\gamma)$, added to the numerical upper endpoint. The exterior remainder enters the lower endpoint. Compressing the combined integral and exterior factors Z to rank r introduces the two-sided bound $2\sigma_1(Z)\sigma_{r+1}(Z) + \sigma_{r+1}(Z)^2$ on their signed Gram matrix. Quadrature refinement and an empirical arithmetic guard are checked separately; the figures are FP64 evaluations, not interval-arithmetic certificates. Refining quadrature order from 10 to 16 and padding from 20 to 24 changes the lower error curves by at most 9.6×10^{-11} .

The remaining analytic slack comes from the interlacing index shift, the normalization bounds ℓ_γ, L_γ , and additive allowances. Small absolute errors can dominate extremely small eigenvalues; the enclosure is valid but need not be informative for every geometry or mode. In this experiment it is informative both for the plotted ratios and for the full error trajectory. Omitting unresolved target energy from the lower curve only weakens it; it is not interpreted as an exact approximation floor.

Target, geometry, and measurements. All displayed data use

$$f(x) = \frac{\sin(2\pi x) + \frac{1}{2} \sin(6\pi x) + \frac{1}{4} \sin(14\pi x)}{\sqrt{21/32}}, \quad x \in [-1, 1]. \quad (21)$$

The total hidden width is $W = 512$, comprising $N = 467$ interior cells and $R = 22$ halo centers on each side, so $h = 2/467$ and $W = N + 2R + 1$. There are 2,048 uniform training midpoints, 4,096 validation points, and 8,192 dense evaluation points. Every reported error is relative output L^2 error on its stated grid, not squared loss or an EMA. The dense grid checks resolution; it is not an untouched generalization test. Frozen GD starts its readout at zero and executes five million full-batch FP64 updates with $\eta = 1/(2\mu_1)$ at each bandwidth. Figure 4A displays training error. Its spectral reference agrees with executed GD within 2.9×10^{-14} .

Joint training optimizes all slopes, centers (through affine biases), and readout parameters for five million updates on five paired seeds. One recipe per optimizer is selected by median final validation error over all five seeds. The selected Adam and GD schedules are full-horizon cosine decays with initial learning rates 0.05 and 0.2, respectively. Frozen Adam uses the same final-validation selection criterion. Panel B shows raw training errors summarized for display; quoted joint and

frozen-Adam endpoint values use the dense grid. Display bins retain extrema, including spikes, rather than replacing the comparison with first tolerance crossings. Panel C measures $h|a_j|$ for learned features $\tanh(a_jx + b_j)$, using the original reference spacing h .

What the joint experiment establishes. Adam’s median scaled RMS slope is 0.0582, and its 99th percentile is 0.2466; the five seeds have 6, 5, 5, 4, 7 features at or above $1/4$. These statistics describe heterogeneous geometry, not an effective uniform bandwidth. The uniform $1/4$ dictionary is an empirical comparison, not a necessary scale for every target or learned dictionary.

Joint Adam continues improving late in training; its median endpoint error falls about 28% between the two- and five-million-update experiments. The result is a precision gap under a large finite budget, not permanent failure. Checked direct fits and an additional two-million-update readout restart on the learned five-million-update features barely improve their error. Multiplying the learned slopes by 4 or 16 at fixed centers worsens the median refitted error to 0.0916 or 0.139, respectively. These interventions change nonnested feature spaces: they support investigating slopes and centers together, not a theorem that increasing slopes always helps. Neither the fixed-kernel proof nor its scalar GD rates apply to evolving features or Adam’s adaptive preconditioner. The next theoretical question is how evolving features acquire a useful geometry.

Reproducibility. The source and numerical provenance for the new spectral figure are `experiments/expD36_frozen_gamma_probe/section34_spectrum_figure.py` and `output/diagnostics/section34_long_horizon/main_5m/spectrum/relative_rates.json`. The existing three-panel figure, full protocols, checks, and evidence links are indexed in `output/diagnostics/section34_long_horizon/README.md`. Both figures reuse completed runs; preparing this note launches no new training experiments.

References

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